
Jommel Rowin S. Sabater

Manila, Metro Manila, 1007 • sabaterjommelrowin@outlook.com • +63 976-308-1953 • <https://github.com/jomzxc>

Education

FEU Institute of Technology

BS in Computer with Specialization in Software Engineering

Manila, Philippines

2022-Present

Technical Skills

Programming Languages: Python, Java, C++, C, JavaScript, SQL, MATLAB, PHP

Web Technologies: HTML, CSS, Bootstrap

Frameworks: Django, FastAPI

Databases: SQLite, MySQL, PostgreSQL

Cloud and DevOps: Google Cloud Platform, Docker, Git, Github

AI/ML: Pytorch, Nvidia NeMo

Projects

Cleft2Speech

Present

Developed a comprehensive web program for the speech rehabilitation of individuals with cleft disabilities. The project involved engineering and deploying an AI-driven API for speech recognition, an adapter to convert transcribed text to ARPABET format, and a user speech cloning API to enable a practice model, alongside training a transducer model to facilitate effective speech therapy.

Project X for NASA '25

2025

Built and deployed a full-stack web application that classifies NASA TESS Objects of Interest (TOIs) into Planet Candidates or False Positives using a custom-trained machine learning model.

Project NATURE for NASA '24

2024

Co-developed an interactive web-based storytelling platform designed to make complex climate change data more accessible and engaging. The project condensed information from the U.S. Greenhouse Gas Center and other sources into easily digestible narratives, allowing users to explore localized stories about global climate change contributions. This platform aims to increase public awareness and encourage personal responsibility through an interactive and narrative-driven educational experience.

Real-time PM2.5 Air Quality Monitoring System

2022

Designed and implemented a full-stack air quality monitoring system. The project involved building an Arduino-based device with a laser sensor to accurately measure PM2.5 concentration. I also developed a complementary Android application for real-time data visualization, allowing users to view the PM2.5 readings from the monitor.

BALAM: Speech-Enabled Virtual Assistant

2021

Developed a comprehensive, Python-based virtual assistant for Windows, featuring hands-free interaction through speech recognition. The program was engineered to enable full PC control via voice commands, including managing applications, system shutdown, and sleep/hibernation. Additionally, it provided real-time data such as time, location, and weather, and enhanced user productivity with music playback and web-based information retrieval.

Leadership & Activities

FEU TECH ASSOCIATION FOR COMPUTING MACHINERY STUDENT CHAPTER

Manila, Philippines

Director for Academics

2023-2024

Lead and streamlined academic initiative, including organizing high-impact review sessions for over 300 students, crafting original problem sets for programming competitions, and developing debate topics and competition guidelines.

FEU TECH ASSOCIATION FOR COMPUTING MACHINERY STUDENT CHAPTER

Manila, Philippines

Junior Officer for Academics

2022-2023

Collaborated on organizing academic initiatives for computer science students, including review sessions and creating

Jommel Rowin S. Sabater

challenging problem sets for programming competitions.